

Quiz 10 Review & Algorithm Deep Dive

Routing Algorithms, OSPF, and BGP

Bernal Manzanilla

Concordia University of Edmonton
IT 213: Networking

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Agenda: Quiz Review

- **Part 1: Routing Algorithm Fundamentals**

- Link-State (LS) vs. Distance-Vector (DV)
- The Adjacency Matrix

- **Part 2: Algorithm Pseudocode**

- Dijkstra's Algorithm (LS)
- Bellman-Ford Equation (DV)

- **Part 3: Scaling with OSPF and BGP**

- Intra-AS Routing (OSPF limits)
- Inter-AS Routing (BGP Attributes & Rules)

- **Part 4: SDN Architecture**

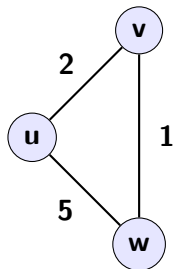
Q1: Link-State vs. Distance-Vector

How do routers map the network?

- **Link-State (LS) Algorithm:** Requires every router to have a complete, global map of the network topology. Routers broadcast their link costs to everyone in the domain.
- **Distance-Vector (DV) Algorithm:** Decentralized. Routers only know the costs to their directly attached neighbors and rely on those neighbors to share their own distance estimates.

Q12, Q13: The Adjacency Matrix (Link-State)

Before running Dijkstra's algorithm, a Link-State router builds a cost matrix of the entire network.



$$\begin{bmatrix} 0 & 2 & 5 \\ 2 & 0 & 1 \\ 5 & 1 & 0 \end{bmatrix}$$

- **Main Diagonal (Q13):** The cost of a router reaching itself $c(x, x)$ is always **0**.
- **No Direct Link (Q12):** If two nodes do not share a physical link, the cost is set to **Infinity** (∞).

Q11: Dijkstra's Algorithm (Link-State)

Computes the shortest path from source u to all other nodes.

```
/* Q11: INITIALIZATION STEP */
```

```
N' = {u}  /* Add starting node 'u' to known set N' */
```

```
for all nodes v
```

```
    if v is a neighbor of u
```

```
        then  $D(v) = c(u,v)$  /* Set cost to direct links */
```

```
    else  $D(v) = \infty$  /* Set others to infinity */
```

```
/* PRIMARY LOOP */
```

```
Loop
```

```
    find w not in N' such that  $D(w)$  is a minimum
```

```
    add w to N'
```

```
    update  $D(v)$  for each neighbor v of w and not in N':
```

```
         $D(v) = \min(D(v), D(w) + c(w,v))$ 
```

```
until N' = N
```

Distance-Vector Algorithm (Bellman-Ford)

Decentralized, iterative, and asynchronous.

Initialization:

for all destinations y in N :

$D_x(y) = c(x,y)$ /* Cost to neighbors; else ∞ */

send distance vector D_x to each neighbor

Loop (forever):

wait (until a link cost changes OR receive DV from neighbor)

for each destination y :

/* The Bellman-Ford Equation: */

$D_x(y) = \min_v \{ c(x,v) + D_v(y) \}$

if $D_x(y)$ changed for any destination y :

send updated distance vector D_x to all neighbors

Q2, Q3, Q14: OSPF and Scaling Limits

- **The Scaling Limit (Q14):** Why not run Link-State across the whole Internet? Building the matrix and running the algorithm scales at an $\mathcal{O}(n^2)$ complexity. The broadcast overhead would instantly overwhelm the global network.
- **The Solution (Q3):** We divide the Internet into Autonomous Systems (AS). An OSPF router only broadcasts its link-state info to other routers **within its own AS**.
- **OSPF Protocol (Q2):** OSPF is the standard intra-AS protocol, and it utilizes Dijkstra's Link-State algorithm to compute paths.

Q4, Q6: Introducing BGP

- **Inter-AS Routing (Q4):** The Border Gateway Protocol (BGP) is the de facto standard for routing *between* Autonomous Systems. It ties the thousands of ISPs together.
- **eBGP vs. iBGP (Q6):**
 - **eBGP** (External): Spans two different Autonomous Systems (router to router across borders).
 - **iBGP** (Internal): Distributes the external routes to all the other routers *inside* the same Autonomous System.

Q5, Q7: BGP Attributes

BGP does not use Link-State matrices. It advertises "Paths."

- **The AS-PATH (Q5):** This attribute lists the Autonomous Systems the advertisement has passed through. Its primary purpose is to **prevent routing loops** (if an AS sees its own number in the path, it rejects it).
- **The NEXT-HOP (Q7):** Identifies the specific IP address of the router interface that begins the AS-PATH in the adjacent Autonomous System. It bridges the gap between intra-AS and inter-AS routing.

Q8, Q10: BGP Route Selection

When a router receives multiple routes to the same destination, how does it choose?

- **Elimination Rules (Q10):** BGP applies rules in order. The very first attribute checked is the **Local Preference** value, which is assigned by the network administrator based on policy.
- **Hot Potato Routing (Q8):** If preferences and path lengths are tied, the router acts selfishly. It routes the packet to the gateway with the **smallest intra-AS cost**. It wants to get the packet out of its own AS as quickly as possible, ignoring external costs.

Q9: Software-Defined Networking (SDN)

- **Traditional Routing:** Routers run OSPF or BGP internally and calculate their own tables.
- **SDN Architecture (Q9):** The control plane is physically separated from the data plane. The physical routers receive a forwarding table computed and pushed down by a **remote, logically centralized SDN controller**.